An Exact Band–Gap Atlas for the Delta-Comb Kronig–Penney Hamiltonian

Route-A source-local certificate HCS-C327

3 September 2026

Abstract

We realize the one-dimensional periodic delta comb by a closed quadratic form for every real coupling and compute its monodromy discriminant without a formal-potential shortcut. Floquet reduction gives an exact spectral criterion, pure absolute continuity, and a complete negative-energy atlas; the coupling $ga = -4$ is the precise zero-energy threshold. This is the operator and discriminant owner.

1 Closed form, matching owner, and Floquet discriminant

Fix a period $a > 0$ and a coupling $g \in \mathbb{R}$. On $L^2(\mathbb{R})$ define

$$\mathfrak{h}_{a,g}[u] = \int_{\mathbb{R}} |u'(x)|^2 \, dx + g \sum_{n\in\mathbb{Z}} |u(na)|^2, \qquad \operatorname{dom} \mathfrak{h}_{a,g} = H^1(\mathbb{R}). \tag{1}$$

The periodized trace inequality makes the sum finite and infinitesimally form bounded relative to the kinetic term. Thus (1) is closed and lower semibounded for either sign of g and defines a unique self-adjoint operator $H_{a,g}$.

Proposition 1 (Operator and monodromy)**.** *The operator acts as $-u''$ off $a\mathbb{Z}$ and has domain characterized by $u \in H^1(\mathbb{R})$, cellwise H^2 regularity with $u'' \in L^2(\mathbb{R} \setminus a\mathbb{Z})$, and*

$$u'(na+) - u'(na-) = gu(na). \tag{2}$$

For $E = k^2 > 0$, propagation from one right limit to the next is conjugate to the determinant-one matrix

$$M(E) = \begin{pmatrix} 1 & 0 \\ g & 1 \end{pmatrix} \begin{pmatrix} \cos(ka) & \sin(ka)/k \\ -k\sin(ka) & \cos(ka) \end{pmatrix}. \tag{3}$$

Its half trace is

$$\Delta(E) = \cos(ka) + \frac{g}{2k} \sin(ka). \tag{4}$$

At $E = 0$ this means $\Delta(0) = 1 + ga/2$ by continuity. At $E = -\kappa^2 < 0$ it means

$$\Delta(-\kappa^2) = \cosh(\kappa a) + \frac{g}{2\kappa} \sinh(\kappa a). \tag{5}$$

Proof. On every centered cell I_n the one-dimensional trace estimate gives, for each $\varepsilon > 0$,

$$|u(na)|^2 \le \varepsilon \|u'\|^2_{L^2(I_n)} + C_{a,\varepsilon} \|u\|^2_{L^2(I_n)}.$$

Summing the disjoint-cell estimates proves the asserted form bound; choosing ε after g proves lower semiboundedness and closedness by the form perturbation theorem. Integration by parts identifies the associated operator and (2). Free propagation followed by the jump gives (3); direct multiplication gives determinant one and (4). Replacing k by $\mathrm{i}\kappa$ gives (5), including the displayed limit at zero. □

The Floquet transform decomposes $H_{a,g}$ over $\theta \in [-\pi, \pi)$ into compact-resolvent fibres on $[-a/2, a/2]$. The fibre functions are quasi-periodic at the endpoints and obey (2) at the origin. A fibre eigenvalue satisfies

$$\Delta(E) = \cos\theta. \tag{6}$$

Theorem 2 (Spectral type and multiplicity)**.** *For every $a > 0$ and $g \in \mathbb{R}$,*

$$\operatorname{spec}(H_{a,g}) = \{E \in \mathbb{R} : |\Delta(E)| \le 1\}, \qquad \operatorname{spec}(H_{a,g}) = \operatorname{spec}_{\mathrm{ac}}(H_{a,g}). \tag{7}$$

There is neither point nor singular-continuous spectrum. The full-line Bloch multiplicity is two in every band interior. If $g \neq 0$, each band edge is a simple periodic or antiperiodic fibre eigenvalue. If $g = 0$, the positive Bragg contacts $(n\pi/a)^2$, $n \ge 1$, are double fibre eigenvalues; $E = 0$ is the simple free bottom.

Proof. Equation (6) and unimodularity give (7). The fibre eigenvalue branches are nonconstant real-analytic functions, with local analytic relabelling at crossings. Their critical points are isolated, so pushing Lebesgue measure in θ through the branches is absolutely continuous; this is also the standard periodic-singular Floquet theorem. Thus their direct integral has neither point nor singular-continuous part. Interior energies have the two distinct angles $\pm\theta$. At an edge the fibre multiplicity is the dimension of $\ker(M \mp I)$. For $g \neq 0$, M cannot equal $\pm I$: at a Bragg point the jump factor is nontrivial, while away from a Bragg point the upper-right entry in (3) is nonzero. Hence the kernel is one-dimensional. For $g = 0$, free propagation equals $(-1)^n I$ at every positive Bragg contact, giving dimension two. The constant periodic mode is the sole zero-energy fibre eigenfunction. □

2 Negative spectrum and the zero threshold

Put

$$q = ga, \qquad z = Ea^2. \tag{8}$$

This scaling removes a from the atlas. For $g < 0$ write $h = -q > 0$ and $y = \kappa a$. Then

$$\Delta_-(y) - 1 = 2\sinh(y/2)[\sinh(y/2) - \frac{h}{2y}\cosh(y/2)], \tag{9}$$

$$\Delta_-(y) + 1 = 2\cosh(y/2)[\cosh(y/2) - \frac{h}{2y}\sinh(y/2)]. \tag{10}$$

The functions $2y\tanh(y/2)$ and $2y\coth(y/2)$ are strictly increasing; their ranges are respectively $(0,\infty)$ and $(4,\infty)$.

Theorem 3 (Complete negative and zero-energy atlas)**.** *If $g > 0$, there is no nonpositive spectrum. If $g = 0$, the spectrum begins at zero. Suppose $g < 0$, and let $y_+ > 0$ be the unique solution*

$$h = 2y_+\tanh(y_+/2). \tag{11}$$

The first band has lower edge $-y_+^2/a^2$, where $\Delta = 1$.

1. *If $0 < h < 4$, zero lies in the interior of this band; its upper edge is positive.*
2. *If $h = 4$ (equivalently $ga = -4$), zero is its simple antiperiodic upper edge, with $\Delta(0) = -1$.*
3. *If $h > 4$, there is a unique $y_- > 0$ satisfying*

$$h = 2y_-\coth(y_-/2), \tag{12}$$

with $y_- < y_+$. The first band is exactly $[-y_+^2/a^2, -y_-^2/a^2]$, and $(-y_-^2/a^2, 0]$ lies in a gap.

Proof. For $g > 0$, (5) is greater than one at every nonpositive energy. For $g < 0$, equations (9) and (10) reduce the two edge equations to (11) and (12). The derivative of the first edge function is positive. After multiplication by $\sinh^2(y/2)$, the derivative of the second has numerator $\sinh y - y > 0$; its left limit is four. The factor signs show that $|\Delta_-| \le 1$ for $0 \le y \le y_+$ when $h \le 4$, and for $y_- \le y \le y_+$ when $h > 4$. Reversing $E = -y^2/a^2$ proves every case. □